UNIFIED UPPER BOUND FOR REGULAR ADDITIVE BASES IN ODD-ORDER RANK-TWO GROUPS

WENKAI YANG

Editorial note. Complete original author mathematical body and embedded bibliography retained. Authors own correctness; no independent proof certification or mathematical repair.

Editorial note. Unused bbm package omitted in derived source0; no mathbbm command occurs in original body. Raw TeX retained.

Editorial note. Unified c0 theorem is counted once, including its decomposition and Kneser arguments; intro split-case consequences not extra targets. N(G) target bundles original lower-bound construction and every necessary cyclic/index-p/quotient lemma.

Editorial note. Original acknowledgement says material is based on parts of author doctoral dissertation completed in2025; this provenance remains in source context.

Allowed standard prerequisites (editorial).

- Finite abelian sumsets, regular/weak-regular sequences and Kneser theory. All original sequence multiplicity conventions, additive-basis and N(G) definitions, quotient and subgroup classification, cited base values and supporting author proofs retained.. Complete original mathematical body and embedded bibliography; prime-minimality and odd-order hypotheses unchanged.

1. Introduction and main results

Let G be a finite abelian group, with additive notation, and let S be a sequence over G. Denote by $\sum(S)$ the set of sums of nonempty subsequences of S. The sequence S is an additive basis of G if $\sum(S) = G$. For a subgroup $H \leq G$, let S_H be the subsequence consisting of all terms of S in H. We call S regular over G if $|S_H| \leq |H| - 1$ for every proper subgroup $H < G$. Let $c_0(G)$ be the least integer l such that every regular sequence over G of length at least l is an additive basis. The problem of determining $c_0(G)$ was proposed by Olson; see [2, 6].

Let p be the smallest prime divisor of $|G|$, and let

$$m(G) = \begin{cases} |G|, & G \text{ is cyclic}, \\ 2p-1, & G = C_p^2, \\ np+2p-3, & G = C_p \oplus C_{np},\ n \geq 2, \\ |G|/p+p-2, & \text{otherwise}. \end{cases}$$

The inequality $c_0(G) \geq m(G)$ is known for every finite abelian group [6, Lemma 2.3]. Equality has been established for many groups of rank at least three; see [2, 6]. For a group $G = C_{n_1} \oplus C_{n_2}$ with $1 < n_1 \mid n_2$, the known cases include the following:

(1) $|G|$ is even, or G is a p-group [2, 10];
(2) $G = C_3 \oplus C_{3q}$, where $q \geq 5$ is prime [11];
(3) $p \geq 3$, $n_1 \geq 2p$ and $|G| \geq 72p^6$ [5];
(4) $p \geq 11$ and $n_1 \geq p^2$ [12].

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Qu, Li, Wang and Zhao also proved that $c_0(G) \le 2|G|/5+1$ for every group of rank two with $p \ge 11$ [12, Lemma 7]. For $G = C_p \oplus C_{np}$ with $n \ge 2$, Qu and Han conjectured that $c_0(G) = np + 2p - 3$ [10, Concluding Remark].

We also study the invariant $\mathsf{N}(G)$ introduced by Gao et al. [3]. A sequence S over G is weak-regular if $\mathsf{v}_g(S) \le \mathrm{ord}(g)$ for every $g \in G$, where $\mathsf{v}_g(S)$ denotes the multiplicity of g in S. The invariant $\mathsf{N}(G)$ is the least integer l such that every weak-regular sequence S over G with $|S| \ge l$ has a nonempty zero-sum subsequence T satisfying $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$. Equivalently, $\mathrm{supp}(ST^{-1}) \ne \mathrm{supp}(S)$. Gao et al. determined $\mathsf{N}(C_p^r) = r(p-1)+1$ for $p \le 3$ and $\mathsf{N}(C_p^2) = 4p - 2\lceil 2\sqrt{p+1}\rceil + 1$ for $p \ge 3$ [4, Theorem 1.3]. For cyclic groups, Jiang and the author proved

$$2n - \lceil 2\sqrt{n+1}\rceil \le \mathsf{N}(C_n) \le 2n - \lceil 2\sqrt{n+1}\rceil + 1 \qquad (n \ge 3),$$

with each equality holding for infinitely many n [8, Theorem 1.1].

Our bounds for additive bases follow from the estimate

$$c_0(C_q \oplus C_{nq}) \le \max\left\{ nq + 2q^2 - q - 1,\ \frac{nq(q+1)}{p} - q + p - 2 \right\},$$

where q is prime, $n \ge 2$, and $p \ge 3$ is the smallest prime divisor of nq; see Theorem 3.5. This estimate applies whether or not $q \mid n$. The first two theorems give consequences for the two cases separately.

Theorem 1.1. *Let q be prime and let $G = C_q \oplus C_{nq^2}$, where $n \ge 1$ is not a power of q. If $p \ge 3$ is the smallest prime dividing $|G|$, then*

$$c_0(G) \le \max\{\frac{nq^2(q+1)}{p}, nq^2 + f(q) - 1\},$$

where $f(q) = 4q(q-1)+1$ for $q \ge 7$, $f(q) = 2q^2+3$ for $q \le 5$.

Theorem 1.2. *Let q be prime and let $G = C_q \oplus C_{nq}$, where $n \ge 1$ and $q \nmid n$. If $p \ge 3$ is the smallest prime dividing $|G|$, then*

$$c_0(G) \le \max\{(2q-1)(n+q+1), \frac{nq^2}{p} + 2\frac{nq}{p}\}.$$

For $\mathsf{N}(G)$, we prove the lower bound $r(n_r - 1) + 1$. The other term in the following theorem is the direct-sum bound of Gao et al. [4, Theorem 1.1(2)].

Theorem 1.3. *Let $G = C_{n_1} \oplus \cdots \oplus C_{n_r}$, where $1 < n_1 \mid \cdots \mid n_r$. Then*

$$\mathsf{N}(G) \ge \max\left\{ r(n_r - 1) + 1,\ \sum_{i=1}^{r} \big(\mathsf{N}(C_{n_i}) - 1\big) + 1 \right\}.$$

Theorem 1.4. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Then*

$$2np - 1 \le \mathsf{N}(G) \le \begin{cases} 2np - 1, & p \mid n, \\ 5np - 4, & p \ge 5,\ p \nmid n, \\ 4np - 3, & p = 3,\ 3 \nmid n. \end{cases}$$

In particular, $\mathsf{N}(C_p \oplus C_{np^2}) = 2np^2 - 1$ whenever $n \ge 1$ and $p \ge 3$ is the smallest prime dividing the group order. The two upper bounds for $p \nmid n$ also hold when $n = 1$.

Section 2 gives the notation and preliminary lemmas. Section 3 proves Theorems 1.1 and 1.2. Section 4 proves Theorems 1.3 and 1.4. The final section discusses the remaining rank-two cases and two cyclic families.

2. Notation and preliminaries

Our notation for sequences follows the standard conventions of zero-sum theory; see [1, 7]. A sequence over G is finite and unordered, and repetitions are allowed. For a sequence $S = g_1 \cdots g_l$, its length is $|S| = l$, its sum is $\sigma(S) = g_1 + \cdots + g_l$, and its support is $\mathrm{supp}(S) = \{g \in G : \mathsf{v}_g(S) > 0\}$. The relation $T \mid S$ means that T is a subsequence of S; then ST^{-1} is the subsequence obtained by deleting T from S. The empty sequence is denoted by 1 and has sum 0. Two subsequences T_1, T_2 of S are disjoint if $T_1T_2 \mid S$. A homomorphism $\varphi : G \to G_1$ induces the sequence $\varphi(S) = \varphi(g_1) \cdots \varphi(g_l)$ over G_1.

Let $\sum_0(S) = \{0\} \cup \sum(S)$. Thus $\sum_0(S)$ includes the sum of the empty subsequence, whereas $\sum(S)$ consists of sums of nonempty subsequences. A nonempty sequence is a minimal zero-sum sequence if it has sum 0 and no proper nonempty zero-sum subsequence. The Davenport constant $\mathsf{D}(G)$ is the least integer l such that every sequence over G of length at least l has a nonempty zero-sum subsequence. We use $\mathsf{D}(C_{n_1} \oplus C_{n_2}) = n_1 + n_2 - 1$ for $1 \le n_1 \mid n_2$ [7, Theorem 5.8.3]. Let $\mathbb{N}$ denote the positive integers. We use the usual floor and ceiling functions $\lfloor x \rfloor$ and $\lceil x \rceil$.

For each subset A of G, denote by $\mathrm{st}(A) = \{g \in G : g + A = A\}$ the stabilizer of A in G. Then $\mathrm{st}(A)$ is the maximal subgroup H of G such that $H + A = A$. We use the following form of Kneser's theorem; see [9, Theorem 4.4].

Lemma 2.1 (Kneser)**.** *Let $A_1, ..., A_n$ be nonempty finite subsets of an abelian group G, and $H = \mathrm{st}(A_1 + \cdots + A_n)$ be the stabilizer. Then*

$$|A_1 + \cdots + A_n| \ge |A_1 + H| + \cdots + |A_n + H| - (n-1)|H|.$$

Lemma 2.2 (Cauchy–Davenport)**.** *Let $A_1, ..., A_n$ be nonempty finite subsets of an abelian group $G = C_p$, where p is prime. Then*

$$|A_1 + \cdots + A_n| \ge \min\{p, |A_1| + \cdots + |A_n| - (n-1)\}.$$

Lemma 2.3. *For every finite abelian group G,*

$$m(G) \le c_0(G) \le |G|.$$

See [6, Lemma 2.3] for the lower bound and [2, Lemma 2.4] for the upper bound.

Qu and Han determined $c_0(G)$ for $G = C_p \oplus C_{p^n}$ as follows.

Lemma 2.4 ([10])**.** *Let $G = C_p \oplus C_{p^n}$, where p is prime and $n \ge 2$. Then*

$$c_0(G) = p^n + 2p - 3.$$

Lemma 2.5 ([2, Lemma 2.3(1)])**.** *Let G be a finite abelian group, and let p be the smallest prime dividing $|G|$. Let S be a regular sequence over G of length $|S| \ge \max\{|G|/p + p - 2,\ \mathsf{D}(G)\}$. If $\sum(S) \ne G$, then $\mathrm{st}(\sum(S)) = \{0\}$.*

We use the cyclic bound

$$\mathsf{N}(C_n) \le 2n - \lceil 2\sqrt{n+1} \rceil + 1 \qquad (n \ge 3)$$

from [8, Theorem 1.1], and the value $\mathsf{N}(C_p^2) = 4p - 2\lceil 2\sqrt{p+1} \rceil + 1$ for $p \ge 3$ from [4, Theorem 1.3(3)]. Also $\mathsf{N}(C_2) = 2$.

Lemma 2.6. *Let $G = C_p \oplus C_{np}$, where $p \ge 3$ is the smallest prime divisor of $|G|$. For $1 < m \mid n$, let $H \le pG$ have order m, and let $k = n/m$. Then*

$$2np - 1 - \big(\mathsf{N}(H) - 1 + (m-1)kp\big) \ge \begin{cases} 2kp - 1, & p \mid k,\ m \ne n, \\ 4kp - 3, & p = 3,\ p \nmid k,\ m \ne n, \\ 5kp - 4, & p \ge 5,\ p \nmid k,\ m \ne n, \\ \mathsf{N}(C_p^2), & m = n. \end{cases}$$

Proof. Since $m \ge p \ge 3$, the cyclic bound gives $\mathsf{N}(H) \le 2m-3$. Therefore

$$\begin{aligned} 2np - 1 - \big(\mathsf{N}(H) - 1 + (m-1)kp\big) &= kp(m+1) - \mathsf{N}(H) \\ &\ge kp(m+1) - 2m + 3. \end{aligned}$$

For $m < n$, subtracting $2kp-1$ from the last expression gives $(kp-2)(m-1)+2 \ge 0$. When $p=3$, subtracting $4kp-3$ gives $(m-3)(kp-2) \ge 0$. When $p \ge 5$, subtracting $5kp-4$ gives $(m-4)kp - 2m + 7 \ge 3m - 13 > 0$. These prove the first three estimates. If $m = n$, then $k = 1$, and the left side is at least

$$(p-2)n + p + 3 \ge p^2 - p + 3 > 4p - 7 \ge \mathsf{N}(C_p^2).$$

The strict inequality follows from $p^2 - 5p + 10 = (p-2)(p-3) + 4 > 0$. □

Lemma 2.7. *Let S be a weak-regular sequence over $G = C_n$, where $n \ge 2$ and $|S| \ge n$. Then either $\sum(S) = G$, or there is a zero-sum subsequence $T \mid S$ such that $\mathsf{v}_g(T) = \mathsf{v}_g(S) \ge 1$ for some $g \mid S$.*

Proof. If $\mathsf{v}_g(S) = \mathrm{ord}(g)$ for some g, take $T = g^{\mathrm{ord}(g)}$. Otherwise $0 \notin \mathrm{supp}(S)$, and we may assume $\sum(S) \ne G$. Since $|S| \ge \mathsf{D}(G)$, $\sum(S) = \sum_0(S)$. Let $H = \mathrm{st}(\sum_0(S))$ and $m = |H|$. If $m = 1$, Kneser's theorem gives $|\sum_0(S)| \ge |S| + 1 > n$. Thus $m > 1$, and $\sum_0(S) \ne G$ gives $m < n$. Since $n = dm$ for some integer $d \ge 2$, Kneser's theorem also gives

$$(|S| - |S_H| + 1)m \le \Big|\sum_0(S)\Big| \le n - m, \qquad |S_H| \ge n - d + 2 \ge 2m \ge \mathsf{N}(H).$$

The middle inequality is equivalent to $(d-2)(m-1) \ge 0$. Apply the definition of $\mathsf{N}(H)$ to the weak-regular sequence S_H; all occurrences in S of the resulting element $g \in H$ belong to S_H. □

Lemma 2.8. *Let S, S' be sequences over a finite abelian group G, and let $H \le G$ satisfy $H \subseteq \sum_0(S)$. Suppose that $T_1 \mid S'$ satisfies $\sigma(T_1) \in H$ and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S') \ge 1$ for some $g \in \mathrm{supp}(S') \setminus \mathrm{supp}(S)$. Then SS' has a zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(SS')$.*

Proof. Choose $T_2 \mid S$ with $\sigma(T_2) = -\sigma(T_1)$ and take $T = T_1T_2$. Since $g \notin \mathrm{supp}(S)$, its multiplicity in T equals that in SS'. □

3. Proof of Theorems 1.1 and 1.2

Lemma 3.1. *Let S be a regular sequence over a finite abelian group G. Then*

$$|\sum_0(S)| \ge \min\{|G|, |S|+1\}.$$

In particular, $\sum_0(S) = G$ if $|S| \ge |G| - 1$.

Proof. Suppose that $|\sum_0(S)| < \min\{|G|, |S|+1\}$. Let $H = \mathrm{st}(\sum_0(S))$ and $m = |H|$. Then $H < G$ and $m \le |S|$, so Kneser's theorem gives

$$|\sum_0(S)| \ge (|S| - |S_H| + 1)m \ge (|S| - m + 2)m \ge |S| + 1,$$

a contradiction. □

Lemma 3.2. *Let $K < H < G$, where $|K| \ge 2$ and $H/K \cong C_q$ for a prime q. Let S be a sequence over $H \setminus K$ that is regular over G. If $|S| \ge 2q-1$, then there exists $T \mid S$ such that $|T| \le q$ and $\sigma(T) \in K \setminus \{0\}$.*

Proof. Take $RR' \mid S$ with $|R| = q$ and $|R'| = q-1$. Since R is regular over H and $|H| = q|K| \geq 2q$, Lemma 3.1 gives $|\sum_0(R)| \geq q+1$. Thus there exist $R_1, R_2 \mid R$ whose sums are distinct and belong to the same coset of K. By Cauchy–Davenport, $\sum_0(R') + K = H$. Choose $R_3 \mid R'$ such that $\sigma(R_1R_3), \sigma(R_2R_3) \in K$. Choose $U \in \{R_1R_3, R_2R_3\}$ with $\sigma(U) \neq 0$. Factor U into subsequences whose images in H/K are minimal zero-sum sequences. Each factor has length at most q, and at least one has nonzero sum, as required. □

Lemma 3.3. *Let $K \leq G$ with $|K| \geq 2$ and $G/K \cong C_q^2$ for a prime q. If S is regular over G and*

$$|S| \geq q|K| + 2q^2 - q - 1,$$

then $S = S_K T_1 \cdots T_k R$, where $k = |K| - |S_K|$, $|T_i| \leq q$, $\sigma(T_i) \in K \setminus \{0\}$ and $\sum_0(R) + K = G$.

Proof. Let $\varphi : G \to G/K$ be the natural homomorphism and let $k = |K| - |S_K|$. Starting with SS_K^{-1}, choose at each step a subgroup of order q containing the largest number of remaining images. Before each of the first k steps, the remaining length is at least $|S| - |S_K| - q(k-1) \geq 2q^2 - 1$, so the chosen subgroup contains at least $2q-1$ terms. Apply Lemma 3.2 in its inverse image to obtain $T_1, \ldots, T_k$. The remaining sequence $R = S(S_K T_1 \cdots T_k)^{-1}$ has length $|R| \geq 2q^2 - q - 1$. Suppose that $\sum_0(\varphi(R)) \neq G/K$. Since $|R| > (q+1)(q-2)$, some subgroup $\bar{H}$ of order q contains at least $q-1$ terms of $\varphi(R)$. For $H = \varphi^{-1}(\bar{H})$, Cauchy–Davenport gives $\sum_0(\varphi(R_H)) = \bar{H}$. It also gives $|RR_H^{-1}| \leq q-2$, since otherwise $\sum_0(\varphi(R)) = G/K$. Thus $|R_H| \geq 2q^2 - 2q + 1$. Working backwards, suppose that $T_{j+1}, \ldots, T_k$ lie in H. Before T_j was chosen, at most $q-2+q = 2q-2$ terms lay outside H, whereas at least $2q^2 - 2q + 1$ lay in H. Distinct subgroups of order q in G/K meet only at zero, and the remaining terms have nonzero images. Thus the subgroup containing the largest number of remaining images is $\bar{H}$, forcing T_j to lie in H. Hence all T_i lie in H, and $|S_H| \geq |S| - q + 2 > |H| = q|K|$, a contradiction. □

Lemma 3.4. *Let p be the smallest prime dividing $|G|$, let $K < G$ with $|K| \geq 2$, and let $l \geq 2$ be an integer. Suppose that S is regular over G and admits a decomposition $S = S_K T_1 \cdots T_k R$, where $k = |K| - |S_K|$, $|T_i| \leq l$, $\sigma(T_i) = t_i \in K \setminus \{0\}$ and $\sum_0(R) + K = G$. If*

$$|S| \geq \frac{|G|}{p} + l\frac{|K|}{p} - l + p - 2,$$

then $\sum(S) = G$.

Proof. Let $W = S_K t_1 \cdots t_k$. Since $|W| = |K| \geq \mathsf{D}(K)$, we have $0 \in \sum(S)$. If W is regular over K, then $\sum_0(W) = K$ and the conclusion follows. Otherwise choose a minimal subgroup $H_1 < K$ with $|W_{H_1}| \geq |H_1| = m$. Then $H_1 \neq \{0\}$ and W_{H_1} is regular over H_1. Let $r = m - |S_{H_1}| - 1$. After reordering, $S_{H_1} t_1 \cdots t_r$ is regular over H_1 of length $m-1$. Its subsequence sums together with 0 therefore cover H_1 by Lemma 3.1. Let $W_0 = S t_1 \cdots t_r (T_1 \cdots T_r)^{-1}$ and $H_2 = \mathrm{st}(\sum_0(W_0))$. Since the T_i are disjoint, replacing each selected occurrence of t_i by T_i shows that $\sum_0(W_0) \subseteq \sum_0(S)$. Together with $0 \in \sum(S)$, this gives $\sum_0(W_0) \subseteq \sum(S)$. The subsequence $S_{H_1} t_1 \cdots t_r$ of W_0 has $\sum_0(S_{H_1} t_1 \cdots t_r) = H_1$, so $\sum_0(W_0)$ is a union of cosets of H_1. Hence $H_1 \leq H_2$. We may assume $H_2 < G$; hence $p \leq |H_2| \leq |G|/p$. Since $r \leq |K|/p - 1$, $|W_0| \geq |S| - (l-1)r$ and $|(W_0)_{H_2}| \leq |S_{H_2}| + r \leq |H_2| - 1 + r$, Kneser's theorem gives

$$|\sum_0(W_0)| \geq (|S| - lr + 2 - |H_2|)|H_2| \geq (|G|/p + p - |H_2|)|H_2| \geq |G|.$$

The last inequality follows from $(|H_2| - p)(|G|/p - |H_2|) \geq 0$, and thus $\sum(S) = G$. □

Theorem 3.5. *Let $G = C_q \oplus C_{nq}$, where q is prime and $n \geq 2$, and let $p \geq 3$ be the smallest prime dividing $|G|$. Then*

$$c_0(G) \leq \max\left\{nq + 2q^2 - q - 1,\ \frac{nq(q+1)}{p} - q + p - 2\right\}.$$

Proof. Let S be regular over G with length at least the stated bound. For $K = qG$, we have $|K| = n$ and $G/K \cong C_q^2$. The first term of the bound allows us to apply Lemma 3.3. The second term is the bound in Lemma 3.4 with $l = q$. Hence $\sum(S) = G$. □

Proof of Theorem 1.1. Apply Theorem 3.5 with nq in place of n. It gives

$$c_0(G) \leq \max\left\{nq^2 + 2q^2 - q - 1,\ \frac{nq^2(q+1)}{p} - q + p - 2\right\}.$$

Since $p \leq q$, the second term is at most $nq^2(q+1)/p$. For the first, $f(q) - 2q^2 + q$ equals $2q^2 - 3q + 1$ when $q \geq 7$ and $q + 3$ when $q = 3, 5$, and is positive in both cases. This proves the stated bound. □

Proof of Theorem 1.2. If $n = 1$, then $G = C_q^2$ and $c_0(G) = 2q - 1$, which is at most $(2q-1)(n+q+1)$. Suppose that $n \geq 2$. Since $p \geq 3$ is the smallest prime dividing $|G|$, Theorem 3.5 applies and gives

$$c_0(G) \leq \max\left\{nq + 2q^2 - q - 1,\ \frac{nq(q+1)}{p} - q + p - 2\right\}.$$

The first is at most $(2q-1)(n+q+1)$, since subtracting $nq+2q^2-q-1$ from that expression gives $(q-1)n+2q > 0$. For the second, $p \leq q$ implies $nq(q+1)/p-q+p-2 \leq nq(q+1)/p$. The latter is at most $nq^2/p + 2nq/p$, because their difference is $nq/p > 0$. Taking the maximum of these two comparisons proves the result. □

4. Proof of Theorems 1.3 and 1.4

Proof of Theorem 1.3. Choose a basis $(x_1, \ldots, x_r)$ with $\operatorname{ord}(x_i) = n_i$ and let $g_0 = x_r$, $g_i = x_r + x_i$ for $1 \leq i < r$. All g_i have order n_r, so $S = \prod_{i=0}^{r-1} g_i^{n_r-1}$ is weak-regular. For a zero-sum subsequence $T = \prod_{i=0}^{r-1} g_i^{a_i}$, the coordinates give $a_i \equiv 0 \pmod{n_i}$ for $i \geq 1$ and $\sum_i a_i \equiv 0 \pmod{n_r}$. Thus also $a_0 \equiv 0 \pmod{n_1}$, and none of the a_i equals $n_r - 1$. This proves $\mathsf{N}(G) \geq r(n_r - 1) + 1$. For completeness, consider the direct-sum bound from [4, Theorem 1.1(2)]. For each i, choose a weak-regular sequence over $\langle x_i \rangle$ of length $\mathsf{N}(C_{n_i}) - 1$ having no nonempty zero-sum subsequence that contains all occurrences of any of its elements. None contains 0, and the direct factors intersect only at 0, so their product in G is weak-regular. In a direct sum, a sequence has sum zero if and only if its sum in every coordinate is zero. A zero-sum subsequence containing all occurrences of some element would therefore have the same property in the corresponding factor, contrary to the choice of the sequences. The product has length $\sum_{i=1}^{r}(\mathsf{N}(C_{n_i}) - 1)$, giving the second bound. □

If $rn_r \geq 2\sum_{i=1}^{r} n_i$, the first lower bound dominates the second, since $\mathsf{N}(C_{n_i}) - 1 \leq 2n_i - 2$.

Lemma 4.1. *Let $G = C_p \oplus C_{np}$, where $n \geq 2$ and $p \geq 3$ is the smallest prime dividing $|G|$. Then*

$$c_0(G) \leq \begin{cases} \lceil (4np-2)/3 \rceil, & p \mid n, \\ \lceil (25np-20)/6 \rceil, & p \geq 5,\ p \nmid n, \\ \lceil (16np-12)/5 \rceil, & p = 3,\ 3 \nmid n. \end{cases}$$

Proof. Suppose first that $p \mid n$. If n is a power of p, then $c_0(G) = np + 2p - 3 \le \lceil (4np-2)/3 \rceil$, since $np \ge p^2 \ge 6p-9$. Otherwise $n \ge p(p+2)$, whence $np \ge p^2(p+2) \ge 6p^2 - 3p - 1$. Thus

$$np + 2p^2 - p - 1 \le \frac{4np-2}{3}, \qquad n(p+1) - 2 \le \frac{4np-2}{3}.$$

Theorem 3.5 proves the first assertion. If $p \nmid n$, then $n \ge p+2$, and the same theorem gives $c_0(G) \le 3np$. For $p \ge 5$, this is at most $\lceil (25np-20)/6 \rceil$; for $p = 3$, it is at most $\lceil (16np-12)/5 \rceil$, since $np \ge 15$. □

Proposition 4.2. *Let $G = C_p \oplus C_{np}$, where p is prime and $n \ge 1$, and let l_0 be a positive integer. Let $r = \lceil l_0/(np-1) \rceil$. If*

$$l_0 - \lfloor l_0/r \rfloor \ge c_0(G),$$

then every regular sequence S over G with $|S| \ge l_0$ has a nonempty zero-sum subsequence T satisfying $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$.

Proof. Since $\langle g \rangle < G$ for every $g \in G$, regularity gives $\mathsf{v}_g(S) \le \operatorname{ord}(g) - 1 \le np - 1$. Hence $|\operatorname{supp}(S)| \ge r$. Choose $g_0 \mid S$ with minimum multiplicity $a = \mathsf{v}_{g_0}(S)$. The sequence $S' = Sg_0^{-a}$ contains none of the occurrences of g_0. Then S' is regular, $a \le \lfloor |S|/r \rfloor$, and

$$|S'| \ge |S| - \lfloor |S|/r \rfloor \ge l_0 - \lfloor l_0/r \rfloor \ge c_0(G).$$

Choose $T' \mid S'$ with $\sigma(T') = -ag_0$. Then $T = T'g_0^a$ has the required property. □

For $n \ge 2$ and $p \ge 3$ the smallest prime dividing $|G|$, let

$$\operatorname{ub}(G) = \begin{cases} 2np - 1, & p \mid n, \\ 5np - 4, & p \ge 5,\ p \nmid n, \\ 4np - 3, & p = 3,\ 3 \nmid n. \end{cases}$$

Corollary 4.3. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Every regular sequence S over G with $|S| \ge \operatorname{ub}(G)$ has a nonempty zero-sum subsequence T such that $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$.*

Proof. Apply Proposition 4.2 with $l_0 = \operatorname{ub}(G)$. The corresponding values of r are 3, 6, and 5, according as $p \mid n$, $p \ge 5$ and $p \nmid n$, or $p = 3$ and $3 \nmid n$. In each case, the required inequality is precisely Lemma 4.1. □

Lemma 4.4. *Let S be a non-regular sequence over a finite abelian group G. Then there exists a proper subgroup $H < G$ such that S_H is regular over H, $|S_H| \ge |H|$, and $\sum(S_H) = H$.*

Proof. Choose $H < G$ of minimum order subject to $|S_H| \ge |H|$. For every proper subgroup $H' < H$, minimality gives $|S_{H'}| \le |H'| - 1$. Thus S_H is regular over H. Since $c_0(H) \le |H|$, we obtain $\sum(S_H) = H$. □

Lemma 4.5. *Let $S = S_H S_1$ with $\sum_0(S_H) = H$ and $\operatorname{supp}(S_1) \subseteq G \setminus H$, and let $\Phi : G \to G/H$ be the natural homomorphism. Suppose that some $\bar{g} \in \operatorname{supp}(\Phi(S_1))$ satisfies $\mathsf{v}_{\bar{g}}(\Phi(S_1)) \ge \operatorname{ord}(\bar{g})$ and that $\Phi^{-1}(\bar{g}) \cap \operatorname{supp}(S_1)$ contains at least two elements. Then S has a nonempty zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S_1$.*

Proof. Let $V \mid S_1$ consist of all terms whose image is $\bar{g}$, and let $n = \operatorname{ord}(\bar{g})$. If $\mathsf{v}_h(V) \ge n$ for some $h \mid V$, let $|V| = kn + t$, where $0 \le t < n$, and take $T_1 = Vh^{-t}$. Then $\sigma(T_1) \in H$, and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$ for every $g \mid V$ with $g \ne h$. Such an element exists by hypothesis. Otherwise, choose $g \mid V$. Since $\mathsf{v}_g(V) < n \le |V|$, there is an n-term subsequence $T_1 \mid V$ containing all occurrences of g. Again $\sigma(T_1) \in H$ and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$. Lemma 2.8 completes the proof in both cases. □

Lemma 4.6. *Let $G = C_p \oplus C_{np}$, where $p \geq 3$ is the smallest prime dividing $|G|$, $n \geq 5$, and $H < G$ is cyclic of order np. Let $S = S_H g_0^b$ satisfy $g_0 \notin H$, $\mathrm{ord}(g_0) = np$, $b \leq np-1$, $|S| \geq 2np-1$, and $|S_H| \geq np$. Suppose that every term of S_H generates H. Choose $h \mid S_H$ with minimum multiplicity a. Let $R = S_H h^{-a}$. If $|R| \leq np-1$ and $a < np/2$, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_h(T) = \mathsf{v}_h(S)$.*

Proof. Let $b = ps_g + t_g$, and $R = \prod_{i=1}^{\ell-1} g_i^{ps_i+t_i}$, where $0 \leq t_i, t_g < p$. First suppose that $a \geq 2p$. As $\ell \leq 1 + (np-1)/a$, convexity of $x + (p-1)(np-1)/x$ on $[2p, np/2]$ gives

$$p\Big(s_g + \sum_i s_i\Big) \geq 2np - 1 - a - \ell(p-1) \geq 3np/2 - 3p \geq np - p/2.$$

Thus $s_g + \sum_i s_i \geq n$. Since $s_g \leq n-1$, choose i with $s_i > 0$. Consider $U = (pg_0)^{s_g} \prod_j (pg_j)^{s_j}$ and delete one occurrence of pg_i. The remaining sequence has at least $n-1$ terms, all generating $pH = pG$, so its $\sum_0$ equals pH by Lemma 3.1. The deleted occurrence of pg_i leaves g_i^p unused in the corresponding subsequences of $Rg_0^{ps_g}$. Since $\sum_0(g_i^p) + pH = H$, we obtain $H \subseteq \sum_0(Rg_0^{ps_g})$.

Now suppose that $a \leq 2p-1$. Then $s_g \geq n-2$ and $|R| \geq (n-2)p + 1 \geq 3p+1$. Take disjoint $U, V \mid R$ with $|U| = |V| = p$. Kneser's theorem gives $|\sum_0(U)| \geq p+1$. Hence two different sums from U have the same image in H/pH. Using $\sum_0(V) + pH = H$, extend both by the same subsequence of V to sums in pH. At least one is nonzero. Thus there is $W \mid UV$ with $\sigma(W) \in pH \setminus \{0\}$. Since pg_0 generates pH and $\sigma(W) \neq 0$, every proper subgroup of pH contains at most one term of $(pg_0)^{s_g}\sigma(W)$, and the trivial subgroup contains none. This sequence is therefore regular over pH and has length at least $n-1$, so its $\sum_0$ equals pH. Also $|RW^{-1}| \geq p+1$, and its image has all terms nonzero in H/pH. Again $H \subseteq \sum_0(Rg_0^{ps_g})$. In either case choose $T_1 \mid Rg_0^{ps_g}$ with $\sigma(T_1) = -ah$ and take $T = T_1 h^a$. □

Lemma 4.7. *Let $G = C_p \oplus C_{np}$, where $n \geq 2$ and $p \geq 3$ is the smallest prime dividing $|G|$, and let $H < G$ be cyclic of order np. Let $S = S_H g_0^b$ satisfy $g_0 \notin H$, $\mathrm{ord}(g_0) = np$, $b \leq np-1$, $|S| \geq 2np-1$, and suppose that S_H is regular over H. If $\mathrm{ord}(h') \leq n$ for some $h' \mid S_H$, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_{h'}(T) = \mathsf{v}_{h'}(S)$.*

Proof. Let $a = \mathsf{v}_{h'}(S)$, $R' = S_H(h')^{-a}$, $s = \lfloor b/p \rfloor$, and $U = R'(pg_0)^s$. Regularity gives $a \leq n-1$, while $s \leq n-1$ and $b - s \leq n(p-1)$. Hence $|U| = |S| - a - b + s \geq np$. Suppose that $\sum_0(U) \neq H$, and let $H_1 = \mathrm{st}(\sum_0(U))$ and $d = |H_1|$. Since $0 \nmid U$, Kneser's theorem implies $H_1 \neq \{0\}$. It also gives

$$|U| - |U_{H_1}| \leq np/d - 2, \qquad |U_{H_1}| \leq d - 1 + s \leq d + n - 2.$$

As $p \leq d \leq n$, $|U| \leq np/d + d + n - 4 \leq 3n - 4 < np$, a contradiction. Thus $\sum_0(U) = H \subseteq \sum_0(R'g_0^{ps})$. Choose $T_1 \mid R'g_0^{ps}$ with $\sigma(T_1) = -ah'$; then $T = T_1(h')^a$ has the required property. □

Lemma 4.8. *Let $H = C_3 \oplus C_{3m}$, where $m \geq 1$ is odd. Every regular sequence R over H of length at least $|H| = 9m$ has a nonempty zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(R)$ for some $g \mid R$.*

Proof. Every multiplicity is at most $3m-1$, so $|\mathrm{supp}(R)| \geq 4$. Choose $g \mid R$ of minimum multiplicity a and let $R' = Rg^{-a}$. Then R' is regular over H and $|R'| \geq \lceil 3|R|/4 \rceil \geq \lceil 27m/4 \rceil$. For $m \geq 5$, Theorem 3.5 gives

$$c_0(H) \leq \max\{3m + 14, 4m - 2\} \leq 6m \leq \lceil 27m/4 \rceil.$$

For $m = 1, 3$, use $c_0(C_3^2) = 5$ and $c_0(C_3 \oplus C_9) = 12$. Thus $\sum(R') = H$. Choose $T' \mid R'$ with $\sigma(T') = -ag$. Then $T = T'g^a$ is zero-sum and $\mathsf{v}_g(T) = \mathsf{v}_g(R)$. □

Lemma 4.9. *Let $H = C_m$, let R be regular over H with $|R| \geq m/2$, and let $H_0 \leq H$ be nontrivial. Then $\sum_0(R) + H_0 = H$.*

Proof. Suppose otherwise and let H_1 be the inverse image in H of the stabilizer of the image of $\sum_0(R)$ in H/H_0. For $k = |H_1|$, we have $2 \le k \le m/2$. Kneser's theorem and regularity give

$$|R| - |R_{H_1}| \le m/k - 2, \qquad |R| \le k + m/k - 3 \le m/2 - 1,$$

contrary to $|R| \ge m/2$. The last inequality follows by convexity of $x + m/x$ on $[2, m/2]$. □

Lemma 4.10. *Let $S = S_H S_1$ be a sequence over an abelian group G, where $H = C_m$, $m \ge 2$, S_H is regular over H, $|S_H| \ge m$, and every multiplicity in S_H is smaller than the corresponding element order. Suppose that pairwise disjoint nonempty subsequences of S_1 have sums forming a sequence U over H. If $\sum_0(U)$ contains a nontrivial subgroup $H_0 \le H$, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_h(T) = \mathsf{v}_h(S)$ for some $h \mid S_H$. In particular, this holds if U has no zero term, is supported in a nontrivial subgroup $K \le H$, and $|U| \ge |K| - 1$.*

Proof. The strict multiplicity bound and $|S_H| \ge m$ ensure that S_H has at least two distinct elements. Choose $h \mid S_H$ of minimum multiplicity a. Then $R = S_H h^{-a}$ has length $|R| \ge m/2$. Lemma 4.9 gives $\sum_0(R) + H_0 = H$. Choose $V \mid R$ with $\sigma(V) + ah \in H_0$. Since $H_0 \subseteq \sum_0(U)$, a subsequence of U has sum $-\sigma(V) - ah$. The corresponding pairwise disjoint subsequences of S_1 form a subsequence T_1 with the same sum. Thus $T = Vh^a T_1$ is zero-sum. Since $h \nmid R$ and $h \nmid S_1$, we also have $\mathsf{v}_h(T) = a = \mathsf{v}_h(S)$. For the last assertion, a nontrivial stabilizer of $\sum_0(U)$ is itself contained in $\sum_0(U)$. If the stabilizer is trivial, Kneser's theorem gives $|\sum_0(U)| \ge |U|+1 \ge |K|$, so $\sum_0(U) = K$. □

Lemma 4.11. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Let $S = S_H S_1$ satisfy $|S| \ge 2np-1$ and $\mathsf{v}_g(S) < \mathrm{ord}(g)$ for every $g \mid S$. Suppose that $H < G$ is cyclic of order m, that S_H is regular over H, and that $m \le |S_H| \le \mathsf{N}(H) - 1$. If G/H is cyclic of order greater than p, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$.*

Proof. By Lemma 3.1, $\sum_0(S_H) = H$. Let $k = np/m > 1$, so that $|G/H| = pk$, and let $\Phi : G \to G/H$ be the natural homomorphism. Since $G/pG \cong C_p^2$ and $(G/H)/p(G/H) \cong C_p$, $H/(H \cap pG)$ has order p. Thus $p \mid m$ and, as H is cyclic, $H \cap pG = pH$. We have $k \ge p$ and $|S_1| \ge 2np - \mathsf{N}(C_m)$.

By Lemma 4.5, we may assume that $\Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$ contains exactly one element whenever $\mathsf{v}_{\bar{g}}(\Phi(S_1)) \ge \mathrm{ord}(\bar{g})$. For each $g \mid S_1$, let $r_g = \mathrm{ord}(\Phi(g))$ and choose $\lfloor \mathsf{v}_g(S_1)/r_g \rfloor$ disjoint copies of g^{r_g}. Let T_0 be their product, U the sequence of their sums, and $R = S_1 T_0^{-1}$. Every term of U is nonzero, since a selected subsequence g^{r_g} satisfies $r_g \le \mathsf{v}_g(S) < \mathrm{ord}(g)$. The choice of T_0 also ensures that $\Phi(R)$ is weak-regular.

Suppose that $R_1 \mid R$ satisfies $\sigma(\Phi(R_1)) = 0$ and $\mathsf{v}_{\bar{g}}(\Phi(R_1)) = \mathsf{v}_{\bar{g}}(\Phi(R))$ for some $\bar{g} \mid \Phi(R)$. Then R_1 contains every term of R whose image is $\bar{g}$. Consequently $T_1 = R_1 T_0$ has sum in H and $\mathsf{v}_h(T_1) = \mathsf{v}_h(S_1)$ for every $h \in \Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$. Lemma 2.8 gives the required subsequence of S. In particular, it suffices to prove $|R| \ge \mathsf{N}(G/H)$.

Suppose first that $m > p$, so $m \ge p^2$. By Lemma 4.10, we may assume that $\sum_0(U)$ contains no nontrivial subgroup of H. Let $U_1 = U_{pH}$ and $U_2 = U U_1^{-1}$, with $a = |U_1|$ and $b = |U_2|$. The same lemma, applied to U_1 with $K = pH$, gives $a \le m/p - 2$. Each term of U_1 is the sum of a selected subsequence of length at most pk. If a selected subsequence corresponding to a term of U_2 had length divisible by p, its sum would lie in $H \cap pG = pH$. Its length therefore divides k and is not divisible by p, so is at most k. Since its length is greater than one, $U_2 \ne 1$ implies $k \ge p + 2$.

If $b \le p - 2$, then

$$|T_0| \le k(pa + b) \le np - (p + 2)k, \qquad |R| \ge m(k - 2) + (p + 2)k \ge 2pk.$$

Here $\mathsf{N}(C_m) \le 2m$, and $m(k-2) \ge k(p-2)$ follows from $m \ge p$ and $k \ge p$. Thus $|R| \ge 2pk \ge \mathsf{N}(C_{pk})$.

If $b \ge p-1$, choose $V \mid U_2$ of length $p-1$. Its image in $H/pH = C_p$ has all terms nonzero, so $\sum_0(V)+pH = H$. The stabilizer of $\sum_0(U)$ is trivial, since otherwise it would be a nontrivial subgroup contained in $\sum_0(U)$. For every $W \mid U$, the identity $\sum_0(U) = \sum_0(W)+\sum_0(UW^{-1})$ gives $\mathrm{st}(\sum_0(W)) \le \mathrm{st}(\sum_0(U)) = \{0\}$. Kneser's theorem therefore yields $|\sum_0(U_1)| \ge a+1$. Since $\sum_0(V)$ meets every coset of pH, $|\sum_0(U_1V)| \ge p(a+1)$. Each remaining term of U_2 increases the number of subsequence sums by at least one. Otherwise that nonzero term would belong to the stabilizer of the preceding sumset, which is trivial. It follows that

$$pa+b+1 \le |\sum_0(U)| \le m-1, \qquad |T_0| \le k(pa+b) \le np-2k.$$

Hence $|R| \ge m(k-2)+2k \ge 2pk$, since $m \ge p^2$, $k \ge p+2$, and $p^2(k-2)-2k(p-1) \ge (p-2)(p^2-2) > 0$. Again $|R| \ge \mathsf{N}(C_{pk})$.

It remains to consider $m = p$. Since $pH = \{0\}$ and every term of U is nonzero, every selected subsequence has length at most k. If $|U| \ge p-2$, then $|\sum_0(U)| \ge p-1$ by Cauchy–Davenport. Choose distinct $h, x \in \mathrm{supp}(S_H)$; such a choice exists because $|S_H| \ge p$ and every multiplicity is less than p. Another application of Cauchy–Davenport gives $\sum_0(U) + \{0, x\} = H$. Since $\sum_0(U) \subseteq \sum_0(T_0)$, there is $T_1 \mid T_0x$ with $\sigma(T_1) = -\mathsf{v}_h(S)h$. Then $T = T_1h^{\mathsf{v}_h(S)}$ is zero-sum and $\mathsf{v}_h(T) = \mathsf{v}_h(S)$. We may therefore assume $|U| \le p-3$.

If $T_0 = 1$, then $R = S_1$ and

$$|R| \ge 2pk - \mathsf{N}(C_p) \ge 2pk - 2p \ge \mathsf{N}(C_{pk}).$$

The last inequality uses $pk \ge p^2$ and the cyclic bound $\mathsf{N}(C_{pk}) \le 2pk - \lceil 2\sqrt{pk+1} \rceil + 1$. Otherwise choose $g \mid S_1$ with $\mathsf{v}_g(S_1) \ge r_g$. Let $t = \mathsf{v}_g(R) < r_g \le k$ and $R' = Rg^{-t}$. By the assumption on the cosets, $\Phi(R')$ contains no occurrence of $\Phi(g)$. It is weak-regular and

$$|R'| \ge 2pk - \mathsf{N}(C_p) - (p-3)k - (k-1) = (p+2)k - \mathsf{N}(C_p) + 1 \ge pk.$$

Apply Lemma 2.7 to $\Phi(R')$. If $\sum(\Phi(R')) = G/H$, choose $R_1 \mid R'$ such that $\Phi(\sigma(R_1)) = -\mathsf{v}_g(S_1)\Phi(g)$. Then $T_1 = R_1g^{\mathsf{v}_g(S_1)}$ has sum in H and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$, so Lemma 2.8 applies. Otherwise there is $R_1 \mid R'$ with $\sigma(\Phi(R_1)) = 0$ and $\mathsf{v}_{\bar{h}}(\Phi(R_1)) = \mathsf{v}_{\bar{h}}(\Phi(R'))$ for some $\bar{h} \mid \Phi(R')$. Since $\bar{h} \ne \Phi(g)$, its multiplicity in $\Phi(R')$ equals that in $\Phi(R)$. The argument for R_1T_0 above completes the proof. □

Lemma 4.12. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Let S satisfy $|S| \ge 2np-1$ and $\mathsf{v}_g(S) < \mathrm{ord}(g)$ for every $g \in \mathrm{supp}(S)$. Suppose that $H < G$ is cyclic of order np, that S_H is regular over H, and that $np \le |S_H| \le \mathsf{N}(H)-1$. Then S has a nonempty zero-sum subsequence T such that $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \in \mathrm{supp}(S)$.*

Proof. Let $S_1 = SS_H^{-1}$, and let $\Phi : G \to G/H \cong C_p$ be the quotient map. Lemma 3.1 gives $\sum_0(S_H) = H$. By Lemma 2.8, it suffices to find $T_1 \mid S_1$ with $\sigma(T_1) \in H$ and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$ for some $g \mid S_1$. Since $n \ge p$ and $np \ge p^2$, the cyclic upper bound gives

$$|S_1| \ge 2np - \mathsf{N}(C_{np}) \ge \lceil 2\sqrt{np+1} \rceil - 1 \ge 2p \ge \mathsf{N}(C_p).$$

If $\Phi(S_1)$ is weak-regular, its length gives a subsequence $T_1 \mid S_1$ such that $\sigma(\Phi(T_1)) = 0$ and $\mathsf{v}_{\bar{g}}(\Phi(T_1)) = \mathsf{v}_{\bar{g}}(\Phi(S_1))$ for some $\bar{g} \mid \Phi(S_1)$. Thus $\sigma(T_1) \in H$, and T_1 contains all occurrences of every element of S_1 whose image is $\bar{g}$. Lemma 2.8 applies.

Otherwise some $\bar{g} \in G/H$ has multiplicity greater than p in $\Phi(S_1)$. If $\Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$ contains at least two elements, apply Lemma 4.5. Suppose that this intersection is $\{g\}$. If S_1 has another element $g_0 \ne g$, then $\sum_0(g^pS_H) = G$, because $g+H$ generates G/H

and $\sum_0(S_H) = H$. Choose $T' \mid g^p S_H$ with $\sigma(T') = -\mathsf{v}_{g_0}(S_1)g_0$. Since $g_0 \nmid g^p S_H$, the sequence $T = T' g_0^{\mathsf{v}_{g_0}(S_1)}$ is a zero-sum subsequence of S with $\mathsf{v}_{g_0}(T) = \mathsf{v}_{g_0}(S)$.

It remains to consider $S_1 = g_0^b$, where $b \ge p$ and $b < \mathrm{ord}(g_0) \le np$. Choose $h \mid S_H$ of minimum multiplicity a, and let $R = S_H h^{-a}$. The strict multiplicity bounds and $|S_H| \ge np$ ensure that S_H has at least two distinct elements, so $|R| \ge |S_H|/2$. If $|R| \ge np$, Lemma 3.1 gives $\sum_0(R) = H$. Choose $T_1 \mid R$ with $\sigma(T_1) = -ah$; then $T = T_1 h^a$ has $\mathsf{v}_h(T) = a = \mathsf{v}_h(S)$. Hence assume $|R| \le np - 1$.

Case 1: $|\mathrm{supp}(R)| \ge 2$. Then $2a \le |R| \le np - 1$ and $b = |S| - |R| - a \ge np - a > np/2$, forcing $\mathrm{ord}(g_0) = np$. If some $h' \mid S_H$ does not generate H, then $\mathrm{ord}(h') \le np/p = n$, so Lemma 4.7 applies. We may therefore assume that every term of S_H generates H. For $n \ge 5$, apply Lemma 4.6. The only remaining possibility is $p = n = 3$. Here Rg_0^b is regular over G: a proper subgroup containing any term of R must equal H, where $|R| \le 8$; any other proper subgroup contains no terms or only b occurrences of g_0, and in the latter case has order at least $\mathrm{ord}(g_0) > b$. Since $a \le 4$, we have $|Rg_0^b| \ge 13 \ge c_0(C_3 \oplus C_9) = 12$. Hence there is $T_1 \mid Rg_0^b$ with $\sigma(T_1) = -ah$, and $T = T_1 h^a$ has the required property.

Case 2: $|\mathrm{supp}(R)| = 1$. Then $R = f^c$ for some $f \in H$, with $a \le c \le np - 1$. Since $2c \ge |S_H| \ge np$ and $c < \mathrm{ord}(f)$, we have $\mathrm{ord}(f) = np$. Let $c = ps_f + t_f$ and $b = ps_g + t_g$, where $0 \le t_f, t_g < p$. If $a \ge (n-1)p$, then $s_f \ge n - 1$ and $s_g \ge 1$. Otherwise $a \le (n-1)p - 1$, so $c + b \ge (n+1)p$ and

$$s_f + s_g \ge \left\lceil \frac{(n+1)p - 2(p-1)}{p} \right\rceil = n.$$

Thus $s_f + s_g \ge n$ in both cases, and $s_f \ge \lfloor n/2 \rfloor \ge 1$. Consider the sequence

$$V = (pf)^{s_f - 1}(pg_0)^{s_g} \quad \text{over } pH, \qquad |V| \ge n - 1.$$

Indeed $pH = pG \cong C_n$ and pf generates this group. Since $g_0 + H$ has order p, we have $p \mid \mathrm{ord}(g_0)$, and the integer $d = \mathrm{ord}(pg_0) = \mathrm{ord}(g_0)/p$ satisfies $d > 1$ and $s_g \le d - 1$. No proper subgroup of pH contains pf; one containing pg_0 has order at least d and contains at most $s_g \le d - 1$ terms of V. Consequently V is regular over pH, and Lemma 3.1 yields $\sum_0(V) = pH$. The terms of V are sums of pairwise disjoint subsequences of $f^{ps_f} g_0^{ps_g}$, leaving p occurrences of f unused. Since $\sum_0(f^p) + pH = H$, we obtain

$$H = \sum_0(V) + \sum_0(f^p) \subseteq \sum_0(f^{ps_f} g_0^{ps_g}) \subseteq \sum_0(f^c g_0^b).$$

Choose $T_1 \mid f^c g_0^b$ with $\sigma(T_1) = -ah$. Then $T = T_1 h^a$ is nonempty and zero-sum, with $\mathsf{v}_h(T) = a = \mathsf{v}_h(S)$. □

Proof of Theorem 1.4. The lower bound for $n \ge 2$ follows from Theorem 1.3. For $n = 1$, both stated upper bounds follow from the known value of $\mathsf{N}(C_p^2)$. Fix p and use strong induction on the integers n whose prime divisors are at least p, with the known value of $\mathsf{N}(C_p^2)$ as the base $n = 1$. Let S be weak-regular over G, with $|S| \ge \mathrm{ub}(G)$. If $\mathsf{v}_g(S) = \mathrm{ord}(g)$ for some $g \mid S$, then $T = g^{\mathrm{ord}(g)}$ is zero-sum and $\mathsf{v}_g(T) = \mathsf{v}_g(S)$. Thus every multiplicity is strictly smaller than the element order; in particular, $0 \nmid S$. If S is regular, Corollary 4.3 applies.

Otherwise choose H by Lemma 4.4. Then H is nontrivial, S_H is regular over H, and $|S_H| \ge |H|$. First suppose $H = C_p \oplus C_{mp}$ is noncyclic. If $m = 1$, the known value of $\mathsf{N}(C_p^2)$ is at most $p^2 \le |S_H|$. If $m \ge 2$ and $p \ge 5$, then $|S_H| \ge mp^2 \ge \mathrm{ub}(H)$, so Corollary 4.3, applied within H, suffices. For $p = 3$, use Lemma 4.8. In each case, $\mathsf{v}_h(S_H) = \mathsf{v}_h(S)$ for every $h \in H$, so the subsequence obtained in S_H also has the required property in S.

We may therefore suppose $H = C_m$ is cyclic. If $|S_H| \ge \mathsf{N}(H)$, apply its definition within H. Otherwise $m \le |S_H| \le \mathsf{N}(H) - 1$. Let $S_1 = SS_H^{-1}$, and let $\Phi : G \to G_1 = G/H$

be the quotient map. If G_1 is cyclic, apply Lemma 4.12 when $|G_1| = p$ and Lemma 4.11 otherwise. These lemmas apply because $|S| \geq \mathrm{ub}(G) \geq 2np - 1$.

Suppose finally that G_1 is noncyclic. Since $G_1/pG_1 \cong G/(H + pG)$ has order $p^2 = |G/pG|$, we obtain $H \leq pG$. Thus $1 < m \mid n$ and $G_1 \cong C_p \oplus C_{kp}$, where $k = n/m < n$. If $\mathsf{v}_{\bar{g}}(\Phi(S_1)) \geq \mathrm{ord}(\bar{g})$ for some $\bar{g} \in G_1 \setminus \{0\}$ and $|\Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)| \geq 2$, Lemma 4.5 applies. Otherwise, for each $g \mid S_1$ with $\mathsf{v}_g(S_1) \geq r_g = \mathrm{ord}(\Phi(g))$, choose $\lfloor \mathsf{v}_g(S_1)/r_g \rfloor$ disjoint copies of g^{r_g}. Let T_0 be their product, U the sequence of their sums, and $R = S_1 T_0^{-1}$ the remaining sequence. Every term of U is nonzero because $r_g \leq \mathsf{v}_g(S) < \mathrm{ord}(g)$, and $\Phi(R)$ is weak-regular. If $|U| \geq m - 1$, Lemma 4.10 applies with $K = H$. Otherwise $|T_0| \leq (m-1)kp$, and

$$|R| \geq 2np - 1 - (\mathsf{N}(H) - 1) - (m-1)kp \geq \mathsf{N}(G_1).$$

The last inequality is Lemma 2.6, using the appropriate induction bound for $k \geq 2$, or the known elementary-group value for $k = 1$. By the definition of $\mathsf{N}(G_1)$, there is $R_1 \mid R$ with $\sigma(\Phi(R_1)) = 0$ and $\mathsf{v}_{\bar{g}}(\Phi(R_1)) = \mathsf{v}_{\bar{g}}(\Phi(R))$ for some $\bar{g} \mid \Phi(R)$. Then $T_1 = R_1 T_0$ has sum in H and contains all terms of S_1 whose image is $\bar{g}$. In particular, $\mathsf{v}_h(T_1) = \mathsf{v}_h(S_1)$ for any $h \in \Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$. Since $\sum_0(S_H) = H$, Lemma 2.8 completes the proof. Replacing n by np proves the asserted equality for $C_p \oplus C_{np^2}$. □

5. Remarks and open questions

We conjecture that $\mathsf{N}(C_p \oplus C_{np}) = 2np - 1$ whenever $n \geq 2$ and $p \geq 3$ is the smallest prime dividing np. The non-regular case in the proof of Theorem 1.4 uses only $|S| \geq 2np - 1$. Hence, for the remaining cases $p \nmid n$, the bound $c_0(C_p \oplus C_{np}) \leq \lceil (4np - 2)/3 \rceil$ would imply the conjecture by Proposition 4.2.

For the cyclic groups below, let $A_{n+1} = \lceil 2\sqrt{n+1} \rceil$ and

$$B_{n+1} = \min\{a + b : a, b \in \mathbb{N},\ ab \geq n + 1,\ \gcd(a, n) = 1\},$$

as in [8]. For the following two families, $\mathsf{N}(C_n)$ attains the upper bound $2n - A_{n+1} + 1$ and is strictly larger than $2n - B_{n+1} + 1$.

Theorem 5.1. *Let $n = 2730\big(1 + 105(2t-1)\,(13t - \frac{13\pm1}{2})\big)$ for a positive integer t; equivalently, $n \in T_2 \cup T_3$, where*

$$T_2 = \{\, 2730(2730t^2 - 2835t + 736) \,\}_{t \geq 1},$$
$$T_3 = \{\, 2730(2730t^2 - 2625t + 631) \,\}_{t \geq 1}.$$

Then

$$B_{n+1} = A_{n+1} + 1 = \lceil 2\sqrt{n+1} \rceil + 1,$$

and

$$\mathsf{N}(C_n) = 2n - B_{n+1} + 2 = 2n - \lceil 2\sqrt{n+1} \rceil + 1.$$

Proof. By [8, Lemmas 5.2 and 4.3], $B_{n+1} = A_{n+1} + 1$ and $\mathsf{N}(C_n) \leq 2n - A_{n+1} + 1$. Let $C_n = \langle e \rangle$.

Suppose first that $n \in T_2$, and let $d = 1365(2t-1) - 52$ and $x = (12n - d)/13$. Then $n = d(d-1) - 26$, $13 \mid d$, and d is odd. Since $d \geq 1313$, we have $(d-1)^2 < n+1 < (d - \frac{1}{2})^2$, and hence $A_{n+1} = 2d - 1$. The identity $n = d(d-1) - 26$ gives

$$13x \equiv -d \pmod{n}, \qquad (d-1)x \equiv -2 \pmod{n}.$$

Every common divisor of x and n therefore divides both d and 2, so $\gcd(x, n) = 1$. As $1 < x < n$, the sequence $S = e^{n-d}(xe)^{n-d+1}$ is weak-regular and has length $2n - A_{n+1}$. If a zero-sum subsequence contains all $n - d + 1$ copies of xe, its multiplicity of e is congruent to $(d-1)x \equiv -2 \pmod{n}$. It must therefore equal $n - 2 > n - d$, a contradiction. If a zero-sum subsequence contains all $n - d$ copies of e, then $13x \equiv -d \pmod{n}$ and $\gcd(x, n) = 1$ force its multiplicity of xe to equal $n - 13 > n - d + 1$, also a contradiction.

Suppose now that $n \in T_3$, and let $d = 1365(2t-1)+53$ and $x = (n-d)/2$. Then $n = d(d-1)-26$, d is even, and $d \equiv 1 \pmod{13}$. In particular, $d \geq 1418$ and $A_{n+1} = 2d-1$ as above. We have

$$2x \equiv -d \pmod{n}, \qquad (d-1)x \equiv -13 \pmod{n}.$$

Every common divisor of x and n divides both d and 13, so $\gcd(x,n) = 1$. Again $1 < x < n$, and $S = e^{n-d}(xe)^{n-d+1}$ is weak-regular with $|S| = 2n - A_{n+1}$. A zero-sum subsequence containing all copies of xe would have $n-13 > n-d$ copies of e, by the second congruence. A zero-sum subsequence containing all copies of e would have $n-2 > n-d+1$ copies of xe, by the first congruence and $\gcd(x,n) = 1$. Thus S has no zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for $g \in \{e, xe\}$. In both cases, the definition of $\mathsf{N}(C_n)$ gives $\mathsf{N}(C_n) \geq 2n - A_{n+1} + 1$, proving the equality. □

Remark 5.2. The construction of Theorem 5.1 relies on the specific relation $n = d(d-1)-26$ together with the arithmetic of 13 modulo d (or $d-1$). For the other two families T_4, T_5 identified in [8, Lemma 5.2] (which satisfy $n = d^2 - 64$ rather than $n = d(d-1)-26$), the analogous construction does not apply directly, and whether $\mathsf{N}(C_n)$ on $T_4 \cup T_5$ attains the upper or the lower bound remains an open question.

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References

1. W. Gao and A. Geroldinger, *Zero-sum problems in finite abelian groups: a survey*, Expo. Math. **24** (2006), 337–369.
2. W. Gao, D. Han, G. Qian, Y. Qu and H. Zhang, *On additive bases II*, Acta Arith. **168** (2015), 247–267.
3. W. Gao, S. Hong, W. Hui, X. Li, Q. Yin and P. Zhao, *Representation of zero-sum invariants by sets of zero-sum sequences over a finite abelian group*, Period. Math. Hung. **85** (2022), 52–71.
4. W. Gao, S. Hong, W. Hui, X. Li, Q. Yin and P. Zhao, *Representation of zero-sum invariants by sets of zero-sum sequences over a finite abelian group II*, J. Number Theory **241** (2022), 738–760.
5. W. Gao, Y. Li, Y. Qu and Q. Wang, *Additive bases of finite abelian groups of rank 2*, Integers **22** (2022), Paper No. A37.
6. W. Gao, Y. Qu and H. Zhang, *On additive bases III*, Acta Arith. **193** (2020), 293–308.
7. A. Geroldinger and F. Halter-Koch, *Non-Unique Factorizations. Algebraic, Combinatorial and Analytic Theory*, Pure and Applied Mathematics, Vol. 278, Chapman & Hall/CRC, Boca Raton, FL, 2006.
8. X. Jiang and W. Yang, *A zero-sum problem related to the max gap of the unit group of the residue class ring*, J. Number Theory **260** (2024), 198–211.
9. M. B. Nathanson, *Additive Number Theory: Inverse Problems and the Geometry of Sumsets*, Graduate Texts in Mathematics, Vol. 165, Springer, New York, 1996.
10. Y. Qu and D. Han, *Additive bases of $C_p \oplus C_{p^n}$*, Int. J. Number Theory **13** (2017), 2453–2459.
11. Y. Qu and Y. Li, *Additive bases of $C_3 \oplus C_{3q}$*, Colloq. Math. **169** (2022), 189–195.
12. Y. Qu, Y. Li, Q. Wang and X. Zhao, *A note on additive bases of abelian groups of rank 2*, Graphs Combin. **41** (2025), Paper No. 24.